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ABSTRACT

Gallium oxide (Ga_2O_3) has attracted great attention for electronic device applications due to its ultra-wide bandgap, high breakdown electric field, and large-area affordable substrates grown from the melt. However, its thermal conductivity is significantly lower than that of other wide bandgap semiconductors such as SiC, AlN, and GaN, which will impact its ability to be used in high power density applications. Thermal management in Ga_2O_3 electronics will be the key for device reliability, especially for high power and high frequency devices. Similar to the method of cooling GaN-based high electron mobility transistors by integrating it with high thermal conductivity diamond substrates, this work studies the possibility of heterogeneous integration of Ga_2O_3 with diamond for the thermal management of Ga_2O_3 devices. In this work, Ga_2O_3 was deposited onto single crystal diamond substrates by atomic layer deposition (ALD), and the thermal properties of ALD-Ga_2O_3 thin films and Ga_2O_3–diamond interfaces with different interface pretreatments were measured by Time-domain Thermoreflectance. We observed a very low thermal conductivity of these Ga_2O_3 thin films (about 1.5 W/m K) due to the extensive phonon grain boundary scattering resulting from the nanocrystalline nature of the Ga_2O_3 film. However, the measured thermal boundary conductance (TBC) of the Ga_2O_3–diamond interfaces is about ten times larger than that of the van der Waals bonded Ga_2O_3–diamond interfaces, which indicates the significant impact of interface bonding on TBC. Furthermore, the TBC of the Ga-rich and O-rich Ga_2O_3–diamond interfaces is about 20% smaller than that of the clean interface, indicating that interface chemistry affects the interfacial thermal transport. Overall, this study shows that a high TBC can be obtained from strong interfacial bonds across Ga_2O_3–diamond interfaces, providing a promising route to improving the heat dissipation from Ga_2O_3 devices with lateral architectures.

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As an emerging semiconductor material with an ultra-wide bandgap (4.8 eV) and a high critical electric field (8 MV/cm), β-Ga_2O_3 has attracted great attention for electronic device applications.[1] However, the thermal conductivity of bulk β-Ga_2O_3 (10–30 W/m K, depending on crystal orientations) is at least one order of magnitude lower than those of other wide bandgap semiconductors such as GaN (230 W/m K), 4H-SiC (490 W/m K), AlN (320 W/m K), and diamond (>2000 W/m K).[2–4] Size effects, doping, and alloying can further reduce the thermal conductivity of Ga_2O_3-based materials, which is an essential thermophysical property that impacts the device behavior and reliability.[4] As seen in other wide bandgap devices, higher device operating temperatures (as a result of the low thermal conductivity in β-Ga_2O_3) will result in faster device degradation and shorter lifetimes. Therefore, efficient thermal dissipation while minimizing device junction temperature is one of the main challenges for β-Ga_2O_3-based devices, especially for high power and high frequency devices where localized Joule-heating results in hot-spots that can lead to degradation and failure.[5] While thermal issues in the use of β-Ga_2O_3 are a major concern, the majority of the characterization of Ga_2O_3 devices has been electrical in nature, and only a few studies on thermal performance and thermal management have been reported.[6–9] Recently, exfoliated Ga_2O_3 nanomembrane field effect transistors (FETs) have been fabricated on a single crystal diamond.[9] Even though the weak van der Waals bonding of the Ga_2O_3 nanomembrane and the diamond substrate leads to a small thermal boundary conductance (TBC) (17 MW/m^2 K), the DC power density in excess of 60 W/mm demonstrated by β-Ga_2O_3 has been achieved on a similarly exfoliated device.[5,9] This result demonstrates the great potential of Ga_2O_3 devices

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TABLE I. Sample structures and thermal properties.

	Al	Ga_2O_3	Diamond surface pretreatment	Ga_2O_3-C TBC	Ga_2O_3 k
Samp1	88 nm	115 nm	N/A	N/A	1.76 W/m K
Samp2	84 nm	29 nm	Ultra-clean	179 MW/m^2 K	1.50 W/m K
Samp3	84 nm	30 nm	Ga-rich	136 MW/m^2 K	1.50 W/m K
Samp4	84 nm	28 nm	O-rich	139 MW/m^2 K	1.52 W/m K

when integrated with a high thermal conductivity substrate, even if this integration approach (mechanical exfoliation and transfer of β-Ga_2O_3) is not readily scalable as a thermal management strategy for β-Ga_2O_3 devices. Thus, a scalable approach for integrating β-Ga_2O_3 with high thermal conductivity diamond is still lacking, which is essential for real-world applications.

In this work, we demonstrate the growth of polycrystalline Ga_2O_3 on diamond and investigate the quality of the resulting films and TBC between β-Ga_2O_3 and diamond substrates. Ga_2O_3 was deposited on single crystal diamond using atomic layer deposition (ALD) with three different diamond surface pretreatments and a reference substrate without pretreatment. The thermal properties of the ALD Ga_2O_3 thin films and the Ga_2O_3–diamond interfaces were measured using time-domain thermoreflectance (TDTR). Additionally, material characterization involving transmission electron microscopy (TEM) analysis and x-ray diffraction (XRD) was performed to understand the structure-thermal property relationship.

Unlike the highly reactive plasma environment of chemical vapor deposited (CVD) diamond growth, ALD growth relies on the sequential self-limiting reactions between surface adsorbed metal organic molecules and oxidizing molecules. Single crystalline (100) diamond substrates were obtained commercially (Element Six, thermal grade) and cleaned using a sequence of treatments intended to remove metal and non-diamond carbon contamination: HNO_3:HCl, HNO_3:H_2SO_4, ultrasonic cleaning with ethanol, and finally an HF etch. The substrates were stored in ethanol and were dried in N_2 immediately prior to transfer into the Ga_2O_3 growth reactor. Thin films (~30–115 nm) were deposited on single crystal (100) diamond substrates in a Fiji 200 G2 reactor at 295 °C using alternating cycles of trimethylgallium (TMG, STREM PURATREM) as the Ga precursor and a remote pure oxygen plasma as the oxidizing source. All samples utilized a turbo pump to drop the pressure in the chamber to 8 mTorr during plasma exposure. Under these conditions, the growth rate was 0.65 Å/cycle. All the samples are undoped.

To measure the thermal conductivity of the Ga_2O_3 film, a thick sample (about 120 nm Ga_2O_3) was grown on a single crystal diamond substrate (Samp1). Since this layer is too thick for our thermal measurement system to be sensitive to the interface TBC, thinner (~30 nm) layers of Ga_2O_3 were grown onto other diamond substrates. In addition, the diamond substrates were given different *in situ* surface pretreatments prior to growth to investigate the effect on TBC. The surface of Samp2 was pretreated with a Ga flashoff process to emulate the one typically used to clean surfaces in MBE. This consists of dosing with TMG to create a gallium sub-oxide at the surface, which is subsequently exposed and removed with a hydrogen plasma pulse. The diamond substrate surface of Samp3 was prepared for growth by super-saturating it with ten consecutive Ga pulses prior to growth, while the surface of Samp4 received ten, 10 s, O_2 plasma pulses *in situ* treatment prior to growth. The details of each sample are summarized in Table I. A layer of Al (~84 nm) was deposited on the sample surface as a transducer for TDTR measurements. The thickness of the Al layer was determined by the picosecond acoustic technique during TDTR measurements. The thickness of the Ga_2O_3 films is determined with TEM.

TDTR is a pump-probe technique to measure the thermal properties of both nanostructured and bulk materials.[10–12] A modulated pump beam (400 nm) heats the sample surface, while a delayed probe beam (800 nm) detects the temperature variation of the sample surface. After being picked up by a photodetector and a lock-in amplifier, the signal is fitted with an analytical heat transfer solution to infer the unknown parameters.[5,13–15] Each of the four samples in this work includes three layers: an Al layer, a Ga_2O_3 layer, and a diamond substrate. The thermal conductivity of a bare single crystal diamond substrate was measured as 2169 W/m K first and then input as a known parameter in the data fitting of the Ga_2O_3-on-diamond samples. There are three unknown parameters: the Ga_2O_3-diamond TBC, the Ga_2O_3 thermal conductivity, and the Al-Ga_2O_3 TBC. The sensitivity of these parameters is shown in Fig. 1(a), which are good for accurate measurements. The heat capacity of Al, Ga_2O_3, and diamond used in the data fitting is from the literature.[5,14,16] The values reported in this work are the average values of four different spots measured on each sample with modulation frequencies of 3.6 MHz and 6.3 MHz. The standard derivations of the measured TBC of each sample are smaller than 8% (eight measurements per sample). The pump and probe diameters are 19.0 μm and 13.3 μm. The experimental data are fitted with an analytical heat transfer solution of the three-layer structure to infer the unknown parameters. The experimentally measured data match very well with the theoretical fitting. The detailed derivation of the analytical heat transfer solution can be found in Refs. 10 and 17.

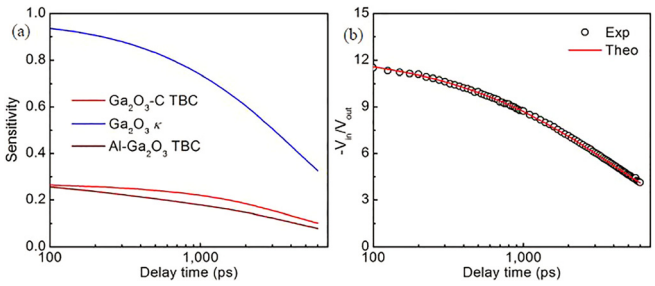


FIG. 1. (a) TDTR sensitivity of the unknown parameters: Ga_2O_3-diamond TBC, Ga_2O_3 thermal conductivity, and Al-Ga_2O_3 TBC. (b) A typical TDTR data fitting (3.6 MHz on Samp2).

03 October 2026 15:56:58

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As shown in Table I, the measured thermal conductivity of Samp1 is 1.76 W/m K, which is much lower than that of bulk β-Ga_2O_3 (10–30 W/m K).[4] To understand this low thermal conductivity value, TEM was used to study the fine structure, as shown in Fig. 2(a). The TEM image of Samp1 shows that the Ga_2O_3 film is polycrystalline with grains on the order of 10–20 nm. The yellow boundaries outline the grains, and the red lines show the orientation of a few of the grains. The strong size effect resulting from small grains limits the phonon mean free path and correspondingly reduces the thermal conductivity of the Ga_2O_3 thin film. Furthermore, some areas are randomly ordered with rough grain boundaries, which result in low phonon transmission and further decrease the thermal conductivity of the Ga_2O_3 thin film. The clean interface of Ga_2O_3 and diamond is shown in Fig. 2(b). The diffraction patterns of Samp1 and another similar sample are shown in Fig. 2(c). The rings of spots in the diffraction pattern confirm that both Ga_2O_3 films are polycrystalline. Theoretical d-spacings of β-Ga_2O_3 were calculated, and the diffraction rings were superimposed on the selected area diffraction (SAD) image. Diamond (111) diffraction points were also observed in the SAD image as shown in the red circle. The diffraction points match with the theoretical diffraction pattern, and each point corresponds to a grain oriented to diffract at that d-spacing. We have also checked the Ga_2O_3 films with XRD, and the typical result from one sample is shown in Fig. 2(d). The peak at 35.2° corresponds to the β-Ga_2O_3 (111) peak, and the peak at 52.6° corresponds to the β-Ga_2O_3 (601) peak.[18] Scherrer's equation has been used to estimate the grain size from the two observed peaks, and the calculated grain size is around 10 nm. This is in good agreement with TEM grain size results and reinforces the assignment of the two peaks with β-Ga_2O_3. Even though there is an overlap of β and ε phase diffraction patterns within this region, there is no plane in the ε phase that corresponds to a peak at 52.6°. The corresponding d spacing of the two peaks shown in the XRD scan is indicated with the yellow circles in Fig. 2(c). This confirms that the films are crystalline β-Ga_2O_3. However, we cannot eliminate the possibility of the existence of ε phases that could reduce the thermal conductivity.

The measured thermal conductivity of Samp2–4 is about 1.5 W/m K, which is much lower than that of the bulk counterpart and slightly lower than that of Samp1. By taking the small grain size (10–20 nm) into consideration, the film thickness (~30 nm) shows a slight thickness dependent thermal conductivity effect, but this effect plays a less important role in limiting phonon mean free paths. As shown in Fig. 3(a), the measured thermal conductivity of Ga_2O_3 thin films is compared with the minimum thermal conductivity of amorphous Ga_2O_3 estimated by two minimum thermal conductivity models: the Cahill model and the diffusion model. The Cahill model is known as the amorphous limit of a material, which assumes the mean-free-paths of the Debye-like, heat-carrying oscillations as half of the corresponding wavelengths. The corresponding minimum thermal conductivity is[13,19]

$$\kappa_{Cahill} = \left(\frac{\pi}{6}\right)^{1/3} k_B n^{2/3} \sum_{i=1}^{3} v_i \left(\frac{T}{\Theta_i}\right)^2 \int_0^{\Theta_i/T} \frac{x^3 e^x}{(e^x-1)^2} dx, \quad (1)$$

where k_B is the Boltzmann constant, v_i is the sound velocity of polarization i, n is the atomic density, and $\Theta_i = v_i(h/2\pi k_B)(6\pi^2 n)^{1/3}$ is the cutoff frequency expressed as temperature unit, where h is the Planck constant.[13] The diffusion model of minimum thermal conductivity introduces a diffusion diffusivity, and the corresponding minimum thermal conductivity is[13,20]

$$\kappa_{diffusion} = \frac{n^{1/3} k_B}{\pi} \int_0^{\infty} \left(\frac{g(\omega)}{3n}\right) \left(\frac{h\omega}{2\pi T k_B}\right)^2 \frac{e^{\frac{h\omega}{2\pi T k_B}}}{\left(e^{\frac{h\omega}{2\pi T k_B}}-1\right)^2} \omega d\omega. \quad (2)$$

The only input in this model is the density of states $g(\omega)$, which is obtained from DFT calculations. We use the single crystal phase to approximate the density of states of the amorphous phase because they are usually similar.[21]

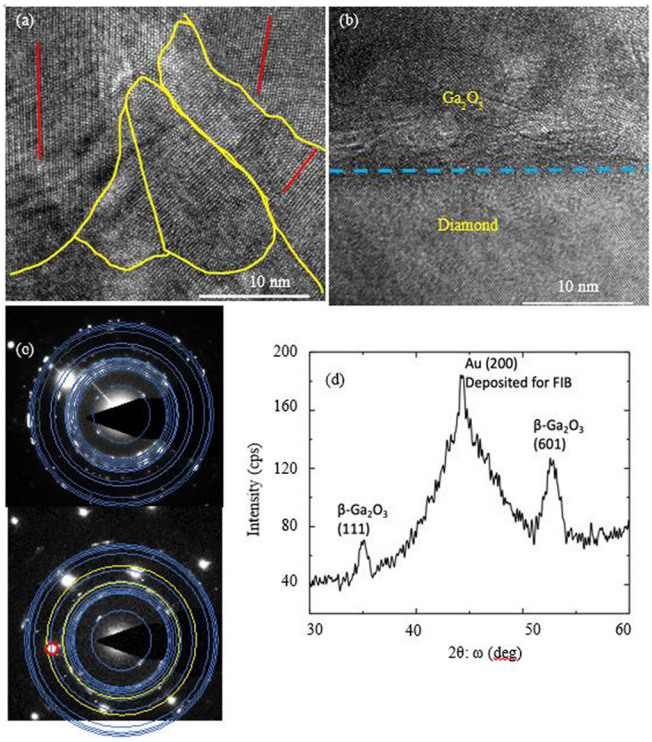


FIG. 2. (a) TEM image of the Ga_2O_3 grain structure of Samp1 and (b) TEM image of the Ga_2O_3–diamond interface of Samp1. (c) The diffraction pattern of Samp1. The spot circled in red is from diamond (111). (d) XRD pattern of Samp1.

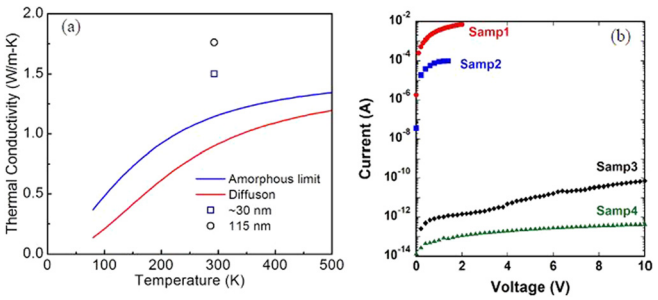


FIG. 3. (a) The comparison of the measured thermal conductivity of Ga_2O_3 thin films with calculated minimum thermal conductivity. The "amorphous limit" is calculated based on the Cahill model of minimum thermal conductivity, while "diffusion" is based on a diffusion model of minimum thermal conductivity.[13,19,20] (b) Current–voltage curves of Samples Samp1–4.

03 October 2026 15:56:58

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The calculated minimum thermal conductivity of Ga_2O_3 is 1.15 W/m K and 0.90 W/m K at room temperature for the amorphous limit and the diffusion model, respectively. Compared to the bulk thermal conductivity (10–30 W/m K), the measured thermal conductivity of the 30-nm-Ga_2O_3 thin films is only 1.3 times as the amorphous limit and 1.67 times as the diffusion minimum thermal conductivity. Crystalline materials usually have a much larger thermal conductivity than their amorphous counterpart. Here, the fine nanocrystalline nature makes the thermal conductivity of the measured films so close to the thermal conductivity of amorphous Ga_2O_3. Figures 4(a) and 4(b) show the TEM images of Ga_2O_3 grown on diamond for Samp2 and Samp4, respectively. All these four samples (Samp1–4) have similar grain structures for these Ga_2O_3 thin films. There are some spotty contrast areas that are about 10 nm away from the Ga_2O_3–diamond interfaces for the thinner films in Figs. 4(a) and 4(b). These areas are possible amorphous or less crystalline regions, which could be further affected or damaged by the focused ion beam (FIB) process during TEM sample preparation, which may contribute to the reduction in the thermal conductivity of Ga_2O_3 thin films. The low thermal conductivity of ALD Ga_2O_3 also highlights the importance of the integration with diamond to extract heat from Ga_2O_3. Additionally, the electrical properties of Samp1–4 are measured, as shown in Fig. 3(b). Circular 500 μm diameter Ti/Au contacts were deposited on the four samples using e-beam evaporation through a shadow mask. No contact annealing was performed. Current–voltage measurements were performed at room temperature using a Keithley 4200SCS semiconductor parameter analyzer. Samp1 and Samp2 exhibited a nearly linear current–voltage relationship, whereas Samp3 and Samp4 were highly resistive. This significant reduction in current could have resulted from improved Ga_2O_3/diamond interface quality after the *in situ* surface pretreatments performed prior to growth, as suggested by the cross-sectional TEM image in Fig. 4(d) for the case of the O_2 *in situ* surface pretreatment. Further characterization of the electrical properties of these films on different substrates will be reported in our future work.

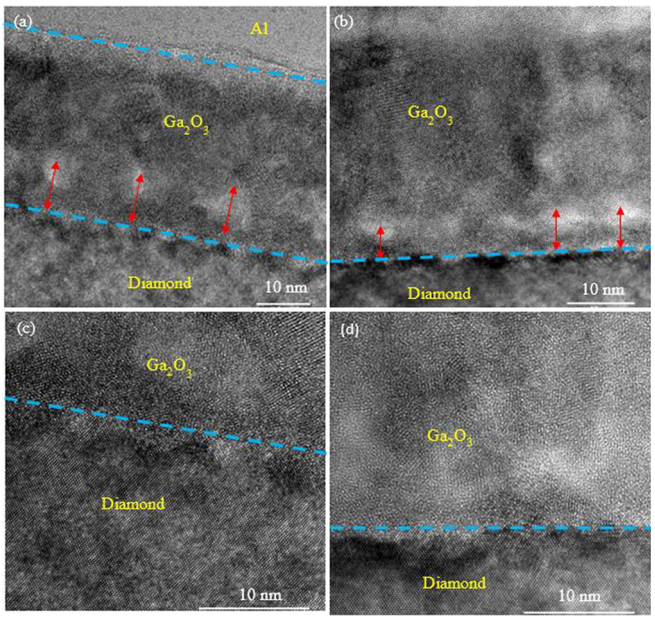


FIG. 4. TEM images of Ga_2O_3 grown on diamond: (a) Samp2 and (b) Samp4. TEM images of Ga_2O_3–diamond interfaces: (c) Samp2 and (d) Samp4.

The higher-resolution TEM images of the Ga_2O_3–diamond interfaces are shown in Figs. 4(c) and 4(d) for Samp2 and Samp4, respectively. The interfaces are not atomically smooth but are atomically abrupt between Ga_2O_3 and diamond without voids or exfoliation at the interfaces. This good contact explains the relatively high TBC of the Ga_2O_3–diamond interfaces reported in Table I. The measured TBC of the ultra-clean (Samp2) interface is 179 MW/m^2 K, about ten times higher than the TBC of a van der Waals bonded Ga_2O_3–diamond interface, suggesting that covalent bonding facilitates interfacial heat transport better than van der Waals interfacial bonding.[5] While the other two thinner samples have smaller TBC due to different *in situ* pretreatments of the diamond surfaces, they are still much larger than that of the van der Waals bonded TBC. This confirms that the type of interface bonding affects TBC significantly, similar to metal–quartz interfaces reported before.[22] Interface chemistry affects the local vibrational modes, resulting in different phonon transport mechanisms at the interface and correspondingly different TBC values. Like epitaxy, the room-temperature surface activated bonding (SAB) technique bonds independently grown layers with covalent chemical bonding interfaces, resulting in high TBC.[16] Thus, from the results shown here, we expect that SAB Ga_2O_3-related interfaces should have high TBC, enabling another possible approach for integrating high-quality β-Ga_2O_3 with high thermal conductivity substrates such as SiC and diamond. Therefore, interface chemistry and bonding are the important factors, which affect the interfacial thermal transport, similar to Al-diamond interfaces reported previously in the literature.[23]

To summarize, Ga_2O_3 is a promising material for electronic device applications, but its thermal conductivity is significantly lower than that of other wide bandgap semiconductors. Thermal management in Ga_2O_3 electronics is the key for device reliability, especially for high power and high frequency devices. This work reported the heterogeneous integration of Ga_2O_3 with diamond for the thermal management of Ga_2O_3 devices. Ga_2O_3 was able to be deposited on single crystal diamond substrates by ALD. TEM studies show small grain sizes and randomly ordered grain boundaries, showing that more work is required to optimize the growth conditions to yield better quality materials. The measured TBC of the Ga_2O_3–diamond interfaces is about ten times larger than that of the van der Waals bonded Ga_2O_3–diamond interfaces, which facilitates thermal dissipation and indicates that interface bonding affects TBC significantly. Good contact between Ga_2O_3 and diamond was observed according to TEM studies. Additionally, the measured TBC of the Ga-rich and O-rich Ga_2O_3–diamond interfaces is about 20% smaller than that of the clean interface, indicating that interface chemistry affects TBC. This study sheds light on the heterogeneous integration of Ga_2O_3 with diamond for the benefit of thermal management of Ga_2O_3 devices, especially for high power and high frequency applications.

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